Group Assignment Week 9

Instructions:

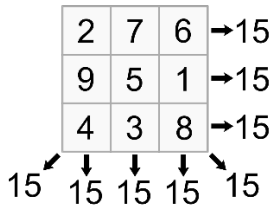
1. Each group can pick any of the 2 problems and do the group assignment.
2. Any one group member can post the code in the discussion. Even though one member posts the code in the discussion, both group members are eligible for 10 points. To see how to post the code in the discussion, refer to the document 'How to post code in D2L Discussions' attached along with the discussion.
3. Code should use the Python commands in the code which are covered in the course till this point of time.

Grading Rubric:

- Code can be compiled and executed = 3 points
- Code complies with assignment instructions = 2 points
- Code is giving expected output = 3 points
- Comments are written in the code = 2 points

Problem1: Lo Shu Magic Square

The Lo Shu Magic Square is a grid with 3 rows and 3 columns as shown below



The grid contains the numbers 1 through 9 exactly.
The sum of each row, each column and each diagonal all add up to the same number.
In a program you can simulate a magic square using a two-dimensional list.
Write a program that declares an empty two-dimensional list. Then the program takes input from the user for numbers and stores in the two-dimensional list.
Assume that the user will enter numbers 1 through 9 exactly once and not repeat any number. Finally, the program should determine whether the list is a Lo Shu Magic Square or not and print it.

Expected Output:

Test Case 1:

```
Enter the numbers for a 3x3 matrix (row by row):
Enter number for position (1,1): 4
Enter number for position (1,2): 9
Enter number for position (1,3): 2
Enter number for position (2,1): 3
Enter number for position (2,2): 5
Enter number for position (2,3): 7
Enter number for position (3,1): 8
Enter number for position (3,2): 1
Enter number for position (3,3): 6

The 3x3 matrix is:
[4, 9, 2]
[3, 5, 7]
[8, 1, 6]

 This is a Lo Shu Magic Square!
```

Test Case 2:

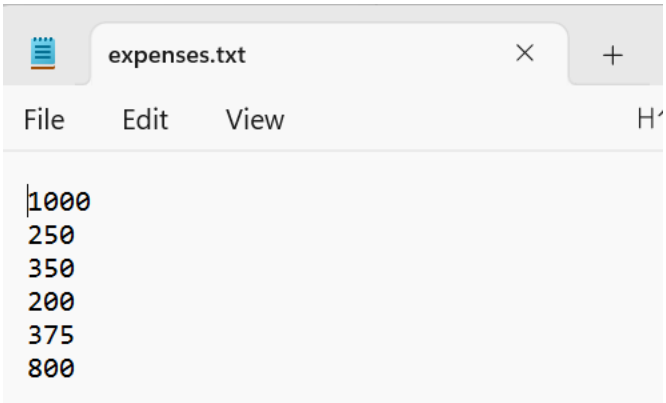
```
Enter the numbers for a 3x3 matrix (row by row):
Enter number for position (1,1): 3
Enter number for position (1,2): 9
Enter number for position (1,3): 2
Enter number for position (2,1): 4
Enter number for position (2,2): 6
Enter number for position (2,3): 7
Enter number for position (3,1): 8
Enter number for position (3,2): 1
Enter number for position (3,3): 5

The 3x3 matrix is:
[3, 9, 2]
[4, 6, 7]
[8, 1, 5]

 This is NOT a Lo Shu Magic Square.
```

Problem 2: Expense Pie Chart

Create a text file named `Expenses.txt` in text editor of your choice. The file contains your expenses for last month in the following categories: [1] Rent [2] Gas [3] Food [4] Clothing [5] Car Payment [6] Misc. Following screenshot is an example of `Expenses.txt`



Write a Python program that reads the data from the file and uses `matplotlib` to plot a pie chart showing how you spend your money.

Expected Output:

